

Experimental Validation of Calcium-Specific Prompt Gamma Energy Lines Induced During Proton Therapy

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Background

- Bone radiotherapy can induce **localized changes in mineral content**, evolving **during treatment** and between fractions. Monitoring these changes for each fraction remains challenging with standard imaging, even though this information would be beneficial for adaptive treatment decisions.^[1]
- Prompt gammas (PG)**: Proton-induced inelastic nuclear scattering populates excited nuclear states, which de-excite via gammas on a ps-ns timescale.
- Prompt Gamma Spectroscopy (PGS)**: extracts compositional data without additional patient dose, previously demonstrated by quantifying oxygen content of irradiated samples via the 6.13 MeV emission line.^[2]
- Current objective**: Identify the optimal energy lines to quantify calcium (Ca). This study experimentally investigates the physical presence and energy dependence of Ca-specific PG emissions and benchmarks them against Monte Carlo simulations.

Materials and Methods

- Experimental Setup**: Irradiations were performed at the Marburg Ion-Beam Therapy Center (MIT). The target was positioned at the isocenter with the detector aligned 34 cm away, shielded by tungsten collimators forming a 12 mm slit to restrict the field of view.
- Detector & Targets**: Spectra were recorded from calcium oxide (CaO) and water targets using a CeBr₃ scintillating crystal surrounded by an active BGO anti-coincidence shield.
- Experiments**: Targets were irradiated with 50 MeV and 100 MeV protons, with a total amount of $\approx 5 \times 10^{11}$ protons. To evaluate energy dependence, spectra were additionally recorded at three different longitudinal positions along the 100 MeV proton beam path.

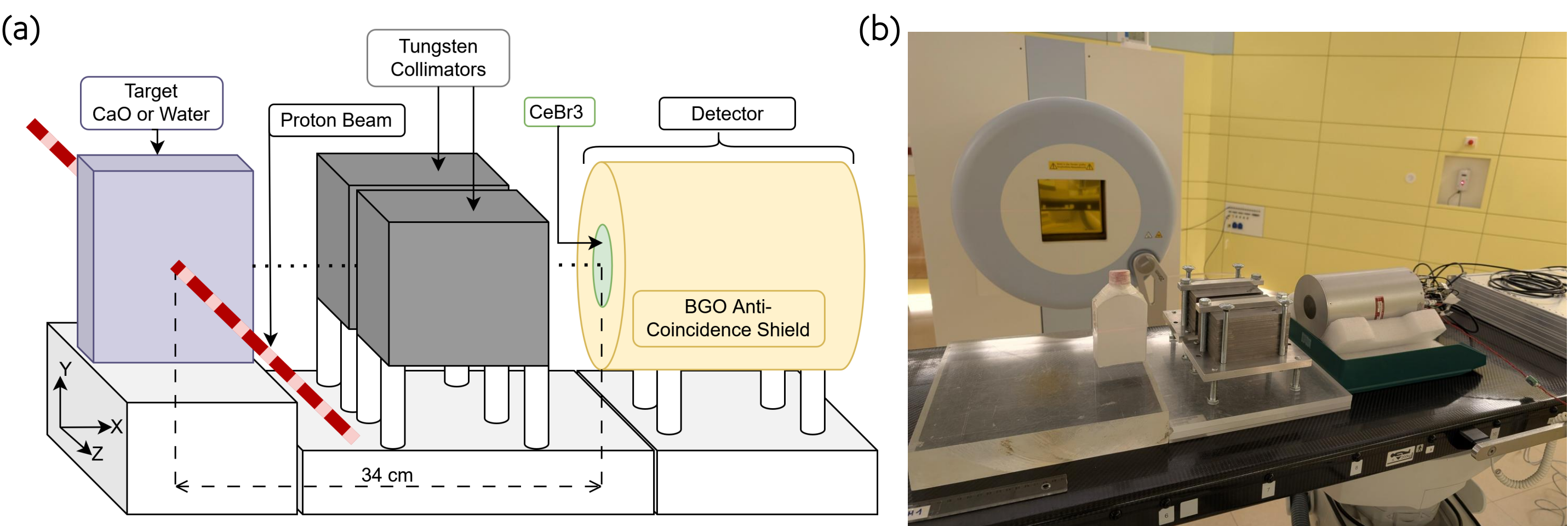


Figure 1: Experimental setup at the MIT beamline. (a) Schematic showing the alignment of the target, tungsten collimators, and the BGO-shielded CeBr₃ detector. (b) Photograph of the physical setup configured for vertical target irradiations.

In Silico Benchmarking & Reaction Identification

- Simulation Benchmarking**: Performed Geant4 simulations that reproduced the experimental spectra. Almost all observed calcium-specific peaks were present in silico and the intensities match.^[3-5]
- Nuclear reaction identification** via Geant4 particle tracking (G4HadronPhysicsQGSP_BIC_HP) cross-checked with the NuDat3 database and TALYS calculations.^[6,7] Multiple nuclear reaction pathways contribute to characteristic spectral peaks.

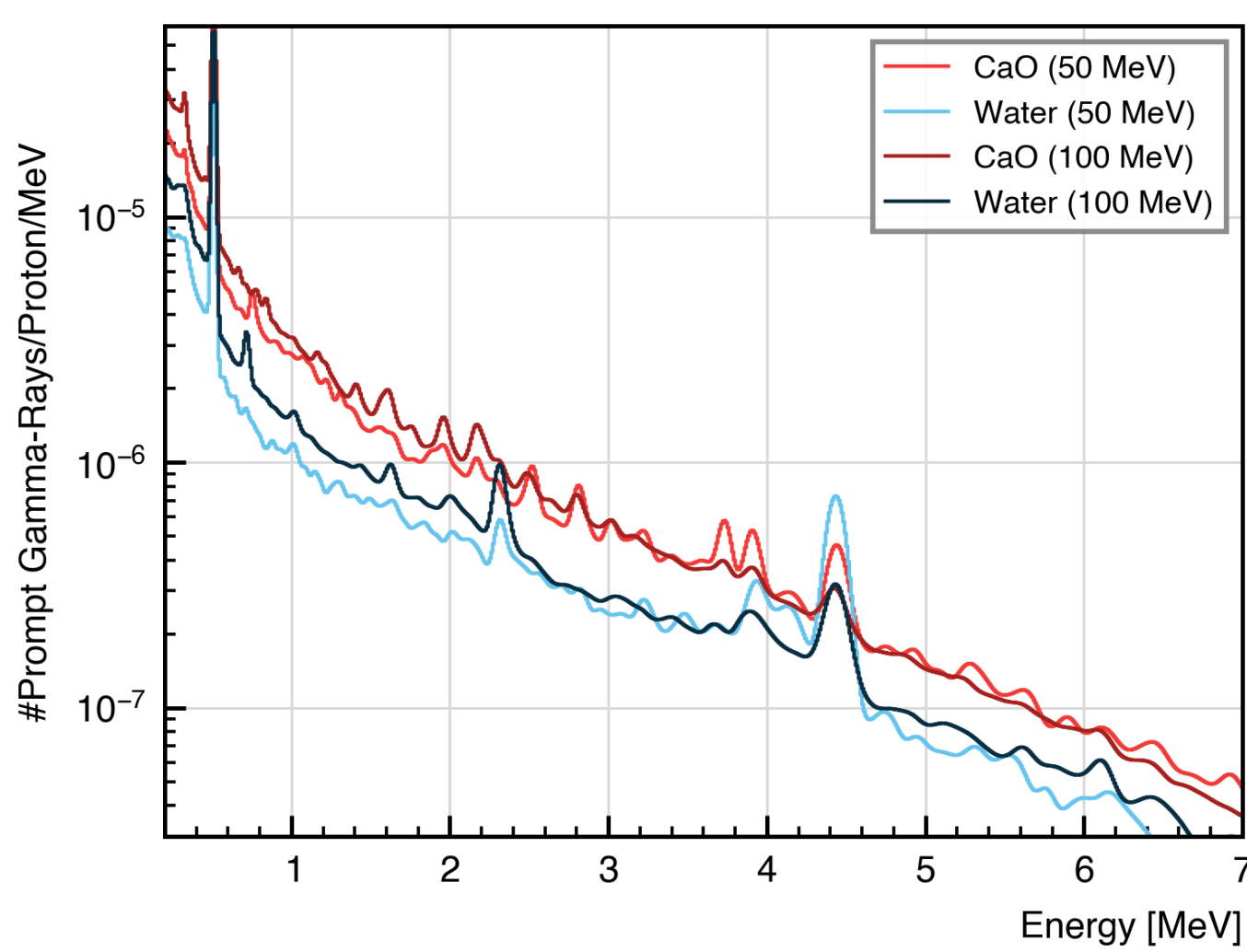


Figure 2: Geant4 simulated spectra reproducing the 50 MeV and 100 MeV proton experiments.

Energy name	Energy [MeV]	Reaction
0.33 (0.33)	0.328	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
0.755	0.755	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
0.77 (0.78)	0.783	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
0.788	0.788	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.165	1.165	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.17 (1.15)	1.219	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.209	1.209	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.374	1.374	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.41 (1.41)	1.410	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.567	1.567	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.611	1.611	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.643	1.643	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.97 (2.00)	1.942	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
1.970	1.970	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.22 (2.22)	2.167	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.217	2.217	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.52 (2.54)	2.467	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.523	2.523	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.81 (2.84)	2.796	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.796	2.796	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
2.814	2.814	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
3.74 (3.76)	3.737	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$
3.90 (3.92)	3.904	$^{40}\text{Ca}_{16}^{20} \rightarrow ^{38}\text{K}_{16}^{20}$

Table 1: Identified prompt-gamma transitions from proton-induced reactions on ^{40}Ca .^[6]

References:

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Results

- Measured PG spectra confirmed the physical presence of multiple calcium-specific prompt gamma lines, which are also observable in the Geant4 simulations.

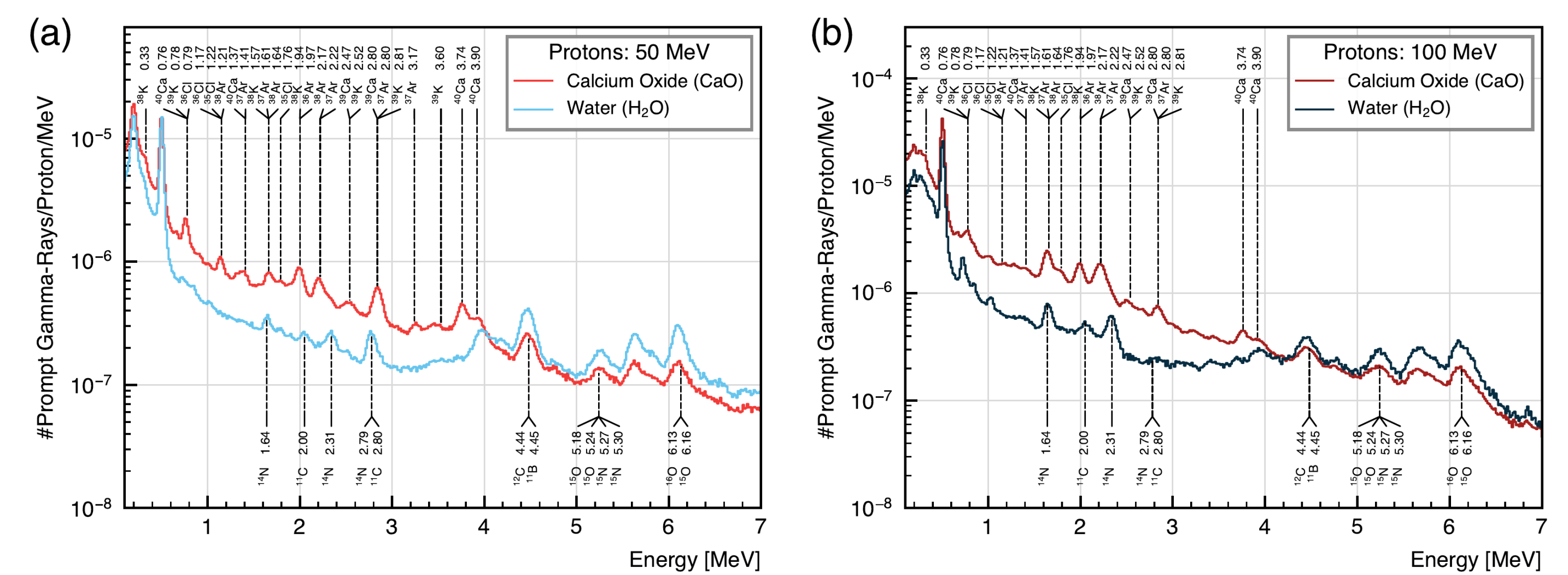


Figure 3: Experimental prompt gamma spectra from calcium oxide (CaO) and water targets irradiated by (a) 50 MeV and (b) 100 MeV protons. Characteristic peaks are annotated with their corresponding product nuclei.

- Energy Dependence**: Emission peak intensities vary across different detector placements along the Bragg curve, aligning directly with the expected energy dependence of inelastic nuclear interaction cross-sections.

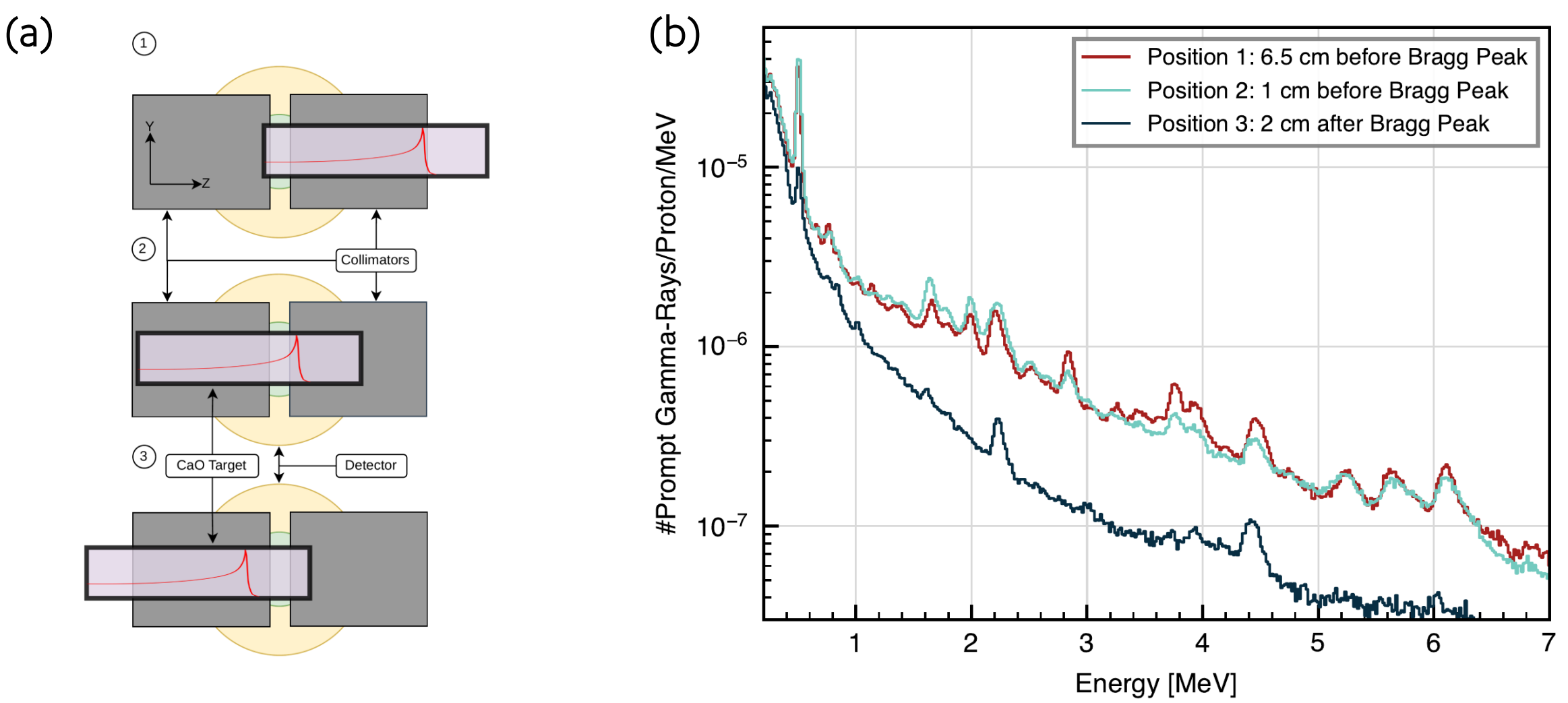


Figure 4: (a) Schematic of detector positions for evaluating depth-dependent emissions. (b) CaO PG spectra measured at these positions with 100 MeV proton irradiation.

Conclusion

- The experimental detection of calcium-specific transition lines validates the physical basis for future PGS applications.
- Depth-dependent variations in measured peak intensities confirm an energy dependence of the underlying inelastic nuclear cross-sections.
- Establishes the foundation for developing prompt-gamma spectroscopy as a clinical tool to monitor bone mineral changes during proton therapy.

Future Work & Outlook

- Build and irradiate complex, tissue-equivalent phantoms** to evaluate calcium-specific spectral features amid background contributions from other biological elements.
- Develop robust analytical methods** to extract relative or absolute calcium mass densities directly from PG spectra.
- Evaluate whether patient geometry and treatment plan data must be integrated** into quantification models to compensate for the observed energy dependence.
- Establish a real-time monitoring method** to provide physicians with direct feedback on bone lesion response during therapy.

Contact and
more
Information:



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